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	Liver diseases affect the lives of millions of Americans. The only available treatment for subjects with end-stage liver disease is liver transplantation. Nationwide, more than 71,000 people are awaiting organ donors, with a new one added every 14 minutes. Last year more than 6,000 patients awaiting organs died, at a rate of 16 a day. However, due to the shortage of donor livers, there is an urgent need to develop alternative approaches, including liver assist devices, xenotransplantation and hepatocyte transplantation. While these approaches require a readily available pool of primary human hepatocytes, previous efforts have failed to establish long-term cultures of viable and differentiated hepatocytes. Our preliminary results using a NASA-developed bioreactor that creates the unique environment of low shear force and enables 3-D cell growth, demonstrated that a simulated microgravity environment is conducive for maintaining long-term cultures of functional hepatocytes. We observed the formation of liver tissue-like structures (up to 3.0 cm in		

length). Albumin mRNA was expressed throughout the 60-day culture. Studies of microsomal functions demonstrated activity of the p450 system, suggesting the feasibility of this approach for pre-clinical metabolic screening of drug candidates. However, significant further expansion of liver mass, or the remodeling of liver in vitro will be implausible without optimizing oxygenation and nutritional uptake within growing cellular aggregates. Our preliminary results also indicated that simulated microgravity modifies the expression of several genes and that microarray technology may provide new understanding of the fundamental biological questions of how gravity affects the development and function of individual cells. Simulated microgravity allows the creation of low shear force, high-mass transfer, and enables 3-dimensional cell growth. Based on our preliminary results, we hypothesized that this unique environment provides outstanding advantages that will allow us to achieve progress in liver tissue engineering. The proposed research is an extension of work that was conducted with the support of NASA. The long-term goal is to establish a basis for a potential use of expanded primary hepatocyte cultures in clinical and commercial applications. The specific objectives of this proposal are:
1) To optimize liver tissue bioengineering in simulated microgravity by evaluating the role of synthetic and biologic scaffolds, co-culturing different cell types derived from liver tissue, and assessing new and evolving bioreactors. 2) To assess molecular mechanism(s) of hepatocyte growth and differentiation in simulated microgravity. 3) To establish an in vivo model for transplantation of bioengineered liver tissue expanded in a microgravity bioreactor.

Task Description:

Liver cells will be harvested from human, pig, and mouse livers by collagenase perfusion and cultured with or without scaffolds in simulated microgravity. Using microarray technology we will assess the effect of microgravity on gene expression in liver cells. Culturing liver cells in a microgravity bioreactor will permit cell differentiation and matrix formation. The resulting cultures will be used for future clinical applications such as hepatocyte transplantation, as an in vitro metabolic model for screening and discovery of new drugs, for the development of extracorporeal liver assist devices, and for gene therapy. In addition, these studies will assist us in establishing an in vivo system, which will support replication of hepatitis viruses and will pave the way for the development of effective antiviral agents. Also, these liver cell cultures will support development of research for the International Space Station and applied research in biotechnology for long-duration space flight. Furthermore, our project will strive to address the request to explore the research related to space flight risks and the CPR critical questions. The long-term goal of this proposal is to establish a basis for a potential use of microgravity conditions in establishing primary hepatocyte culture that will be used for hepatocyte transplantation, development of extracorporeal liver assist devices, and gene therapy. The specific objectives of this proposal are: 1) to optimize liver tissue bioengineering in simulated microgravity by evaluating the role of synthetic and biologic scaffolds, co-culturing different cell types derived from liver, and assessing new and evolving bioreactors; 2) to assess molecular mechanism(s) of hepatocyte growth and differentiation in simulated microgravity; and 3) to establish an in vivo model for transplantation of bioengineered liver tissue expanded in microgravity bioreactor. Millions of Americans suffer tissue or organ loss from diseases and accidents every year, and the yearly cost of treating these patients exceeds $400 billion. The major medical treatment for these losses is transplantation of tissues and organs. Nationwide, more than 71,000 people are awaiting organ donors, with a new one added every 14 minutes. Last year more than 6,000 patients awaiting organs died, at a rate of 16 a day. Unlike other organ failure modalities, orthotopic liver transplantation (OLT) is the only established successful treatment for liver failure. Several approaches are being developed either to provide a bridge to OLT or to reestablish liver function without OLT: transplantation of autologous or allogeneic hepatocytes, ex vivo or in vivo liver-directed gene therapy, xenotransplantation, extracorporeal bioartificial liver assist devices, and tissue engineering of replacement liver tissue. Recent advances in cell biology, tissue culture techniques, and biotechnology have led the way for the potential use of isolated hepatocytes in treating an array of liver disorders. Isolated hepatocytes may be transplanted to replace liver-specific deficiencies or as an important element of an auxiliary hybrid, bioartificial extracorporeal liver support device, which are important therapeutic applications for treating severe liver disease. Progress in these areas has been restrained by the inability to establish the "ideal" hepatocyte in culture. The ideal hepatocyte should be of human origin, normal (nonmalignant) phenotype, readily available, rapidly and easily grown in cell culture, should remain stable in a well-differentiated state for days to weeks, and should be capable of the full range of synthetic and detoxifying features of mature hepatocytes. Previous work has failed to establish a human hepatocyte culture with these properties. Therefore, alternative research models have used primary hepatocytes from other mammalian species (pig, rat, and dog) and immortalized human liver cell lines. Pigs are physiologically close to humans, and their organs are of similar size. Although infection from pigs is viewed as less likely than from baboons, pathogens known as porcine endogenous retroviruses, have raised concerns. They are harmless in pigs, but their effect in humans is unknown. Recently, scientists from BioTransplant revealed data that pig retroviruses has been genetically altered so that they cannot reproduce in humans, effectively rendering the disease harmless. Engineered tissue using the patients' own cells, or possibly immunological inactive allogeneic or xenogeneic cells, has the potential to overcome current problems of replacing lost tissue function and to offer new therapeutic options for diseases with metabolic deficiencies where currently there is no therapy available. These new approaches will include improved systems for extracorporeal artificial liver support and hepatocyte transplantation. Furthermore, our project will strive to address the request to explore the research related to space flight risks and the CPR critical questions.

Rationale for HRP Directed Research:

Research Impact/Earth Benefits:

Our preliminary results using a NASA-developed bioreactor that creates the unique environment of low shear force and enables 3-D cell growth, demonstrated that a simulated microgravity environment is conducive for maintaining long-term cultures of functional hepatocytes. However, significant further expansion of liver mass, or the remodeling of liver in vitro will be implausible without optimizing oxygenation and nutritional uptake within growing cellular aggregates.
Specific aims 1 and 2.

a) Our previous results have demonstrated that simulated microgravity may induce apoptosis in the central zone of large cell aggregates. To investigate the impact of stress-mediated apoptosis in simulated microgravity and to optimize cell culture condition and reduce apoptosis in cells cultured in the rotary cell culture system (RCCS), we are continue to study the effect of ursodeoxycholic acid (UDCA) on human hepatoma cell line, Huh7 and primary pig hepatocytes. It well documented that administration of UDCA, a hydrophilic non-toxic bile acid, improves clinical and biochemical conditions in several liver diseases. UDCA inhibits apoptosis by interrupting classic pathways of apoptosis resulting from various apoptotic stimuli such as deoxycholic acid (DCA), ethanol, okadaic acid, TGF-b, and Fas ligand. UDCA can prevent mitochondrial depolarization in cells and subsequently inhibit cytochrome c release and caspase activation.

Apoptosis is a result of interplay between proapoptotic and survival signals. Therefore, an understanding of antiapoptotic cascade is of importance to prevent toxic/ischemic cellular injury. In a previous report we demonstrated that treatment of cells with UDCA prevented ER-mediated stress and inhibited caspase-12 activation. Now we are expanding our studies regarding the antiapoptotic effects of UDCA.

Survivin is a recently described member of the inhibitors of apoptosis (IAP) family. To date, six human IAPs have been described: NIAP, cIAP1, c-IAP2, XIAP, survivin and BRUCE. Among these IAPs, survivin is unique because it is exclusively over-expressed in transformed cell lines and in human cancer tissues but it is almost undetectable in terminally differentiated adult tissues after fetal development. Expression of survivin correlates with a worsening prognosis in patients with human hematologic and solid cancers such as lymphoma, leukemia, gastric cancer, colorectal cancer, bladder cancer, and neuroblastoma. In vitro, survivin prevents apoptosis in cells exposed to diverse apoptotic stimuli and regulates the cell cycle in the G2/M phase. Survivin, like other IAPs, appears to suppress apoptosis by binding and inhibiting the terminal effectors, caspase-3 and caspase-7. Although the overexpression of survivin can reduce apoptosis induced by growth factor withdrawal or by TNF-a, there is no information available regarding the activity of survivin in ER stress-mediated apoptosis.

Therefore, to assess the activity of survivin and to evaluate the effect of UDCA on the survivin following ER stress-mediated apoptosis, we established a model of thapsigargin (TG)-induced apoptosis in two liver-derived cell lines. TG is a highly selective inhibitor of sarcoplasmic/ER Ca2+-ATPase (SERCA) that disrupts intracellular calcium homeostasis and induces ER stress-mediated apoptosis. ER stress-mediated apoptosis was confirmed by demonstrating the morphological changes of apoptosis, DNA fragmentation, PARP cleavage, activation of calpain, and cleavage of procaspase-12 in both Bcl-2 negative Huh 7 cells and Bcl-2 positive HCC-T cells. This study demonstrated that the level of survivin decreased gradually after TG treatment in both cell lines, indicating that survivin might play a role in ER stress-mediated apoptosis. Although the downstream targets of activated caspase-12 are not clear, our experiments showed that caspase-3 and caspase-7 proteins were activated in the ER stress-mediated apoptotic process. These observations suggest that the two caspases may also be the final common executors of apoptosis as in the other two classic pathways of apoptosis. Therefore, it is conceivable that survivin is involved in TG-induced apoptosis because it binds to caspase-3 and caspase-7 and inhibits their functions in the apoptotic process.

Following these observations we assessed mitochondrial and ER-stress mediated pathway in microgravity environment. Primary pig hepatocytes and human cell line (Huh7) cultured in simulated microgravity were examined for markers of ER-stress and apoptosis. We observed decrease in ER stress following reduction of calcium efflux, inhibition of Bip/GRP78, and caspase-12 activation. Inhibition of activation of NFkappaB, a regulator of anti-apoptotic genes, was also detected. Subsequent studies demonstrated inhibited activation of effector caspases and apoptosis. However, the levels of PARP and pro-caspases-9 and -3 were significantly higher in cells cultured in RCCS bioreactor indicating anti-apoptotic effect of simulated microgravity. In addition to these findings these results were confirmed by flow cytometry when we measured mitochondrial membrane potentiality and annexin V. To assess effects of simulated microgravity on the cell cycle we evaluated the levels of the proliferating cell nuclear antigen (PCNA). PCNA is one of several essential factors for proper chromatin assembly in regard to chaperoning newly synthesized histones to replicating DNA, and its expression is closely associated with cell cycle progression. The levels of PCNA in cells cultured in microgravity bioreactor were 3 fold lower than the controls. Interestingly, levels of PCNA were returned to normal when cells were cultured in medium supplemented with 100 mM ATP or in the presence of PGA-scaffolds.

In summary, we demonstrated that culturing cells in RCCS has cytoprotective effects similar to UDCA and suggest that microgravity environment prevents apoptosis by blocking calcium mediated apoptotic pathway and reduction of ER stress-mediated cell injury. Our results suggest that microgravity has several physiological influences including stimulating anti-apoptotic and anti-proliferative effects.

b) It has been reported that psychiatric issues including anxiety, depression, psychosis, and psychosomatic symptoms have affected astronauts during space flights. Steps can be taken to minimize the impact of these issues, both before and during the mission. The mechanism of space flight-induced depression remains unclear. Recent observations reported that depression has been linked to the activation of the limbic-hippocampal system and dysregulation of the negative feedback system of the hypothalamic-pituitary-adrenal (HPA) axis. Glucocorticoid receptors (GR), located in the limbic-hippocampal area, are key elements in this negative feedback system. GR are also involved in termination of the stress response. Dysregulations and dysfunctions GR have been implicated in the pathogenesis of the depressive disorders. Recently, the serotonin (5-HT) system was discovered to be involved in development of depression. To gain insight into mechanisms of development of depression in astronauts, we assessed the effect of simulated microgravity on the modulation of GR and 5-HTR1A expressions. This aim strives to address the CPR request to evaluate critical questions for space flight risk 28: Human Performance Failure Due to Neurobehavioral Problems. Primary pig hepatocytes were cultured in RCCS bioreactor for 1, 2, and 3 days and analyzed by Western blot and Flow cytometry. Cells cultured in dishes served as controls. No significant changes were observed in 5-HTR1A expressions. Time-dependent decreases of mRNA levels of GR were observed (~50%). A recent report demonstrated that UDCA, which is a cytoprotector, can up-regulate GR expression. Thus, we assessed the effects of simulated microgravity and 100 mM UDCA on GR. Treatment with 300 mM deoxycholic acid (DCA), which is an apoptosis inducer, was a control for UDCA. Our results revealed that the effects of simulated microgravity were decreased when cells were treated with UDCA. Co-incubation with DCA increases the effects of simulated microgravity and further downregulates GR. Interestingly, levels of GR were return to normal when cells were cultured in a medium supplemented with 100 mM ATP or PGA-scaffolds. Our results suggest that microgravity-induced down-regulation of GR expression may be the causative factor in the development of depression in astronauts.

The work accomplished for this project is partially published in the journal Cancer Letters 2003, 191:83-92.

Specific aim 3.

Research in human developmental biology has led to the discovery of human stem cells (precursor cells that can give rise to multiple tissue types), including embryonic stem (ES) cells, embryonic germ (EG) cells, fetal stem cells, and adult stem cells.

Recent evidence suggests that adult-derived stem cells, like their embryonic counterparts, are pluripotent and that stem cells can prompt a damaged organ to regenerate. These simple, undifferentiated and uncommitted cells are able to respond to signals from their host tissue microenvironment and differentiate, producing progeny that display a phenotype characteristic of the mature cells of that tissue. We used a clonal stem cell line (termed WB-F344, gift from Dr. W.B.Coleman) that was derived from an adult male rat liver to study the effects of simulated microgravity on this stem

Task Progress:

cell line. The initial microarray evaluation was performed during early stages of cell assembly in the RCCS that create environment of low shear force without any demonstrable effect of compromised nutrition and oxygen supply in the center of spheroids. Total cell RNA was extracted from WB-F344 cells cultured in the RCCS at the time of the formation spheroids (3 days). Cells cultured in T-75 flask were used as a control. To assess a difference in gene expression in WB-F344 cells cultured in 3-D and 2-D, we used the Affymetrix GeneChip Array at Baylor College of Medicine Microarray Core Facility. The Rat Genome U34A chip contains probe sets interrogating more than 24,000 mRNA transcripts and EST clusters from the UniGene database. Arrays were scanned using a GeneChip Scanner 3000 System (Affymetrix) and analyzed with GeneSpring (Silicon Genetics). Analysis of Affymetrix GeneChip revealed an upregulation of 1,344 genes and a downregulation of 762 genes in cell cultured in simulated microgravity.

The upregulation of the following genes is of interest: Mitochondrial 3-hydroxy-3-methylglutaryl CoA synthase (43-fold) is mediates an early step in cholesterol synthesis. This synthase is the first enzyme of ketogenesis, a metabolic pathway that provides lipid-derived energy for organs during times of carbohydrate deprivation, such as fasting.

The airway trypsin-like protease (15-fold) was found elevated in patients with chronic airway diseases. Glutamate-ammonia ligase (11-fold) plays an important role in controlling body pH and in ammonia detoxification. The enzyme also clears L-glutamate, the major neurotransmitter in the central nervous system, from neuronal synapses. OAZ (9-fold) is a multizinc finger protein, which plays an important role during development of bone, neural and immune systems.

Heme oxygenase (7.5-fold), an essential enzyme in heme catabolism, cleaves heme to form biliverdin, which is subsequently converted to bilirubin by biliverdin reductase, and carbon monoxide, a putative neurotransmitter. Heme oxygenase is an antioxidant that plays a protective role in wound healing and its expression increases during inflammation. Aquaporin-CHIP was up 6.8-fold. This protein is important during development. In transgenic mice with overexpression of mineralocorticoid receptor developed enlarged kidneys associated with renal tubular dilation and cellular vacuolization. These mice also developed mild dilated cardiomyopathy accompanied by a significant heart rate increase.

Serum/glucocorticoid-regulated kinase (SGK) was increased at least 5-fold. SGK is involved in anti-apoptotic effect of glucocorticoid receptor. Furthermore, SGK alterations have also a big effect on cellular metabolism and gene expression in response to osmotic changes, glucose concentrations and oxidative stress. Interferon regulatory factor-1 (4.6-fold) is a transcriptional activator for the interferon-alpha and the interferon-beta genes. Dysfunction or genetic mutations of this gene lead to the most frequent cytogenetic abnormalities in human leukemia and the preleukemic myelodysplastic syndromes.

GAP junction protein (CX26) expression was elevated 5-fold. CX26 expression is regulated by the cell cycle, showing moderate expression during G1 and S and strong upregulation in late S and G2. The IL18 expression was also increased up to 4.5 times. It has been demonstrated that IL18 has a protective effect against the development of chronic graft-versus-host disease in mouse. IL18 also elevated the expression of vascular cell adhesion molecule-1.

The expression of vascular endothelial growth factor (VEGF) was elevated 3.5 times. VEGF is the only mitogen that specifically acts on endothelial cells. Its expression is upregulated by hypoxia, and its cell-surface receptor, known as fetal liver kinase-1 (Flk1) in mice, is exclusively expressed in endothelial cells. The Flk1/VEGF system plays a central role in angiogenesis in general and in the development of solid tumors in particular. Sterol carrier protein-2 (4-fold) is a protein that appears to play an important role in intracellular movement of cholesterol and possibly other lipids. Sterol carrier protein-2 is present in low levels in Zellweger syndrome, in which the cells are deficient in peroxisomes.

Xanthine dehydrogenase (4-fold) is the group of molybdenum-containing hydroxylases involved in the oxidative metabolism of purines. The potentially protective physiologic function of xanthine dehydrogenase in providing urate as an antioxidant defense is the ability of the oxidase form of the enzyme to generate superoxide anion or hydrogen peroxide from molecular oxygen.

Metallothioneins (4-fold) have been postulated to detoxify metals; to play a role in zinc and copper homeostasis during development; to regulate synthesis, assembly, or activity of zinc metalloproteins; and to protect against reactive oxygen species. Fibrinogen (3.7-fold) and fibrin play important, overlapping roles in blood clotting, fibrinolysis, cellular and matrix interactions, inflammation, wound healing, and neoplasia.

Matrix Gla protein (MGP) is a vitamin K-dependent protein initially isolated from bovine bone, was upregulated 3-fold. MGP is expressed at high levels in heart, kidney, and lung and is upregulated by vitamin D in bone cells. Dysfunction of MGP plays a central role in Keutel syndrome, which is an autosomal recessive disorder, characterized by abnormal cartilage calcification, peripheral pulmonary stenosis, and midfacial hypoplasia.

Following genes were down regulated by simulated microgravity condition:

Follistatin was down regulated more than 90-fold. Follistatin-deficient mice are retarded in their growth, have decreased mass of diaphragm and intercostal muscles, shiny taut skin, skeletal defects of the hard palate and the thirteenth pair of ribs, and abnormal whisker and tooth development. They fail to breathe, and die within hours of birth.

A-kinase anchor protein 5 (AKAP) was decreased at least 69 fold. AKAP is important in regulating the activity of T lymphocytes and in determining their sensitivity to incoming cAMP signals.

St2 gene expression was downregulated up to 41-fold. St2 is related to the growth of cells and to control cell proliferation by binding of specific ligand(s). Cyclin D1 gene, involve in cell cycle, and was down 6.4-fold. Cyclin D1 gene can also function as an oncogene. Most human breast cancers overexpress cyclin D1, a component of the core cell cycle machinery. Cyclin D1 -/- mice are resistant to breast cancers induced by the Neu and Ras oncogenes. However, animals lacking cyclin D1 remained fully sensitive to other oncogenic pathways of the mammary epithelium, such as those driven by Myc or Wnt1. Furthermore, other cell cycle related genes such as cell cycle controller (CDC2) and cyclin B were down 4.7- and 3-fold, respectively. Transgelin was downregulated 9.3-fold. This protein is involved in replicative senescence and its downregulation may therefore be an early and sensitive marker for the onset of transformation.

Phosphoribosylpyrophosphate synthetase was downregulated at least 5.8-fold. This gene catalyzes the phosphoribosylation of ribose 5-phosphate to 5-phosphoribosyl-1-pyrophosphate, which is necessary for the de novo and salvage pathways of purine, pyrimidine, and pyridine biosynthesis.

	Endothelin-1 (12-fold) is a potent, 21-amino acid vasoconstrictor peptide produced by vascular endothelial cells. In addition to its vasoconstrictor action, endothelin has effects on the central nervous system and on neuronal excitability. Plasminogen activator inhibitor1 was down 5-fold. This protein involved in wound healing. The expression of enolase was 8 times less then in controls. Downregulation of enolase is associated with exercise intolerance and myalgias. Severe muscle enolase deficiency (5% of residual activity) was detected in this individual, who was found to be compound heterozygous for 2 ENO3 gene mutations. Bone morphogenetic protein 4 (BMP4) was downregulated at least 7.7 times. BMP4 is a vital regulatory molecule that functions throughout development in mesoderm induction, tooth development, limb formation, bone induction, and fracture repair. It is over-expressed in patients with fibrodysplasia ossificans progressive. Interferon-related developmental regulator 1 was down regulated 5.6 times. In addition, we used a stem cell line (WB-F344) to investigate the possibility of the development an in vivo model for transplantation of bioengineered liver tissue expanded in microgravity bioreactor. WB-F344 cells were cultured in RCCS with or without scaffolds (<2x2x2mm) made from polyglycolic acid (PGA) or from collagen. After 7 days in RCCS culture, a tissue-like formations or cell suspension (as a control, cells cultured on plastic dish) were transplanted into subcapsular space of the kidney of mice (strain 129SV male, 10-12 months) by using spinal thin wall T46LNR 18g x 1 ½" B-D needle. After one week the mice were sacrificed and pieces of kidneys fixed in 10% formalin. Thick sections of tissue were analyzed by light microscopy. Histological examination (H/E) revealed that WB-F344 cells survived after transplantation with both scaffolds. Transplantation of cell cultured in the plastic dish was not successful. Immunohistochemical analysis of those experiments is now in progress. The work accomplished for this specific aim is accepted for the poster presentation at the 2005 In Vitro Biology Meeting which will be held from June 5 - 7, 2005 in Baltimore, Maryland.
Bibliography Type:	Description: (Last Updated: 06/07/2007)
Abstracts for Journals and Proceedings	Khaoustov V, Risin D, Pellis NR, Yoffe B. "Liver tissue engineering in microgravity environment." Society for In Vitro Biology meeting, Baltimore, MD, June 5-7, 2005. In Vitro Biology Meeting Abstracts 2005 Jun;61-A. http://www.sivb.org/2005A_p1.pdf (search for Yoffe) , Jun-2005
Articles in Peer-reviewed Journals	Sohn J, Khaoustov VI, Xie Q, Chung CC, Krishnan B, Yoffe B. "The effect of ursodeoxycholic acid on the survivin in thapsigargin-induced apoptosis." Cancer Lett. 2003 Feb 28;191(1):83-92. PMID: 12609713 , Feb-2003